Backend Development Fundamentals

Node.js + Express.js + MongoDB

10-Day Recorded Practical Course - Curriculum Proposal

Duration	Delivery	Approach	Project
10 Days	Recorded	Explain -> Build -> Test	Product CRUD API

Course Purpose

Koorsadani waxay baraysaa aasaaska backend development-ka iyadoo la adeegsanayo Node.js, Express.js iyo MongoDB. Mawduucyada waxaa loo habeeyey si isku xiran; cashar kasta wuxuu dhisayaa aqoonta iyo code-ka casharkii ka horreeyay.

- Fahamka backend, server, API iyo request/response flow.
- Dhisidda Express server iyo REST routes.
- Kala saaridda Route, Controller iyo Model.
- Ku xiridda MongoDB gudaha server.js iyo isticmaalka Mongoose.
- Dhisidda CRUD API oo dhammeystiran iyo Postman testing.
- Image upload iyadoo la adeegsanayo Multer iyo uploads/ folder.
- Soo bandhigidda product data iyo CRUD actions ee simple table interface.

Learning Sequence

```
Backend Concepts -> Node.js -> Express -> Server -> Routes -> Controllers -> MongoDB -> Models -> CRUD -> Postman -> Multer -> Image Upload -> Product Table
```

Teaching & Recording Method

1. Concept-ka marka hore si kooban oo cad ayaa loo sharxayaa.
2. Sharaxaadda kadib isla markiiba practical code ayaa la sameynayaa.
3. Code-ka waxaa lagu run/test gareynayaa terminal ama Postman.
4. Natiijada waxaa lagu xiriirinayaa casharkii hore ka hor inta aan loo gudbin casharka xiga.

```
Explain -> Build -> Test -> Connect -> Continue
```

Habkan wuxuu ka dhigayaa course-ka mid practical ah, isla markaana ardaygu si joogto ah u fahmayo halka qayb walba kaga jirto backend flow-ga.

10-Day Course Structure

Day	Topic	Main Content
Day 1	Backend & Node.js Fundamentals	Backend vs frontend, client/server, API concept, Node.js, npm, package.json, first Node application.
Day 2	Express.js & Server Setup	Install Express, create server.js, app, port, express.json(), first endpoint and server run.
Day 3	Routes & HTTP Methods	GET, POST, PUT, DELETE, request/response, route params, basic Postman testing.
Day 4	Project Structure	Create routes/, controllers/ and models/. Connect Route -> Controller and keep server.js focused.
Day 5	MongoDB Fundamentals	Database, collection, document, MongoDB setup and connection inside server.js.
Day 6	Mongoose Model & Schema	Create Product schema/model with name, price, description, image and status fields.
Day 7	CRUD Part 1 - Create & Read	Create product, get all products, get one product, API responses and Postman tests.
Day 8	CRUD Part 2 - Update & Delete	Update product, delete product, complete CRUD flow and error checks.
Day 9	Multer & Image Upload	Create upload middleware, uploads/ folder, multipart/form-data, store image path in MongoDB.
Day 10	Final Integration & Product Table	Connect complete API flow and display products in a simple table with Add, Edit and Delete actions.

Course Continuity

Course-ku wuxuu ku socdaa hal project oo sii kordhaya. Server-ka la sameeyo bilowga ayaa lagu darayaa routes, controllers, models, database operations, CRUD iyo image upload ilaa final integration-ka.

Project Architecture

```
backend-course/
|-- models/
|   `-- productModel.js
|-- controllers/
|   `-- productController.js
|-- routes/
|   `-- productRoutes.js
|-- middleware/
|   `-- upload.js
|-- uploads/
|   `-- product-images
|-- server.js
|-- .env
`-- package.json
```

MongoDB connection-ka waxaa lagu haynayaa server.js si structure-ku u ahaado mid fudud oo diiradda saaraya fundamentals-ka.

Backend Request Flow

```
Postman / Frontend -> Route -> Multer (when file exists) -> Controller -> Model -> MongoDB -> Response
```

Product Data Model

Field	Type	Purpose
name	String	Product name
price	Number	Product price
description	String	Product description
image	String	Uploaded image path
status	String / Boolean	Active or inactive state

CRUD API Endpoints

Method	Endpoint	Operation
POST	/api/products	Create product
GET	/api/products	Read all products
GET	/api/products/:id	Read one product
PUT	/api/products/:id	Update product
DELETE	/api/products/:id	Delete product

Image Upload with Multer

Image file-ka actual-ka ah wuxuu galayaa uploads/ folder, halka MongoDB lagu kaydinayo path-ka file-ka.

```
File location: uploads/laptop.jpg
MongoDB field: image = "uploads/laptop.jpg"
```

Final Product CRUD Interface

Final integration-ka waxaa lagu muujinayaa simple product table si loo arko API data-ga iyo CRUD actions-ka.

No.	Image	Name	Price	Description	Status	Edit	Delete
1	[IMG]	Laptop	$500	HP Laptop	Active	Edit	Delete

Interface-ka waxaa sidoo kale ku jiri doona + Add Product button oo loo isticmaalo Create operation-ka.

Expected Course Outcome

- Learner-ku wuxuu sharxi karaa backend request flow-ga.
- Wuxuu abuuri karaa Express server iyo REST API routes.
- Wuxuu kala saari karaa Route, Controller iyo Model responsibilities.
- Wuxuu ku xiri karaa Node.js application MongoDB.
- Wuxuu dhisi karaa oo test-gareyn karaa complete CRUD API.
- Wuxuu isticmaali karaa Multer si uu image u upload-gareeyo.
- Wuxuu API data ku soo bandhigi karaa simple CRUD table.

Suggested Recording Format

Section	Suggested Duration
Lesson goal / introduction	1-2 min
Concept explanation	3-7 min
Practical coding	10-20 min
Run / Postman test	3-5 min
Recap / connection to next lesson	1-2 min

Separate Post-Course Module: AI Concepts & Practical Usage

Important: Qaybtan AI-ga waa module madax-bannaan oo la bixinayo kadib marka 10-ka maalmood ee Backend Development Fundamentals la dhammeeyo. Ma aha qayb ka mid ah backend chain-ka aasaasiga ah.

Module Purpose

Module-kan wuxuu learner-ka siinayaa faham guud oo AI ah iyo hab sax ah oo AI tools loogu isticmaalo software development-ka. Ujeeddadu ma aha in AI lagu beddelo fahamka code-ka, balse in loo adeegsado assistant waxbarasho iyo development support.

Topics Covered

- AI waa maxay? Fahamka guud ee Artificial Intelligence iyo Generative AI.
- AI assistant waa maxay, maxaase uu qaban karaa marka coding la sameynayo?
- Sida prompt cad loo qoro: context, task, expected output iyo constraints.
- Sida AI loogu weydiiyo sharaxaad code iyo concept.
- Sida AI loogu isticmaalo debugging iyo fahamka error messages.
- Sida AI looga caawisto API examples, sample JSON iyo test cases.
- Sida AI-generated code loo akhriyo, loo hubiyo loona tijaabiyo ka hor isticmaalka.
- Xadka AI: khaladaadka suuragalka ah, verification iyo muhiimadda developer judgment.

Practical AI Exercises

```
Prompt 1: "Explain this Express route and show how the request reaches the controller."

Prompt 2: "I received this error in my Node.js API. Explain the cause and show the smallest safe fix."

Prompt 3: "Create sample JSON data for testing this Product API in Postman. Do not change my project structure."
```

Course Scope Note

Authentication, authorization, advanced pagination/filtering, DevOps deployment iyo complex cloud image storage kuma jiraan 10-day fundamentals course-kan. Mawduucyadaas waxaa loo diyaarin karaa Backend Intermediate ama Full Stack Development course.